

**Before the
NATIONAL SCIENCE FOUNDATION
Alexandria, VA 22314**

In the Matter of	)	
	)	
Request for Information on the Development	)	
of an Artificial Intelligence (AI) Action Plan	)	Document No. 2025-02305
	)	

COMMENTS OF LUMEN¹

Lumen Technologies, Inc. (Lumen) is pleased to respond to the National Science Foundation’s (NSF) Request for Information to provide input to the Development of an Artificial Intelligence (AI) Action Plan (Plan) as directed by Presidential Executive Order on January 23, 2025. Lumen applauds the President’s commitment to policies that will sustain and enhance U.S. global dominance in AI in order to promote human flourishing, economic competitiveness and national security.

Lumen also agrees with the President’s observation that with the right government policies, innovation, and entrepreneurial spirit that the U.S. is capable of winning the global AI race.² For instance, China's long-term AI strategy, launched in 2017, sets ambitious goals for AI as an economic driver and for China as a global AI hub.³ Recent announcements from

¹ This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

² National Science Foundation, Request for Information on the Development of an Artificial Intelligence (AI) Action Plan, 90 Fed. Reg. 9088 (Feb. 6, 2025).

³ According to the World Economic Forum (WEF), China’s long-term AI strategy, underpinned by adaptive regulations and robust infrastructure, exemplifies how to balance innovation with safety for sustainable transformation. The 2017 “Next Generation AI Plan” aims to position AI as a core driver of economic transformation by 2025 and establish the country as a global hub for

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Chinese AI startup DeepSeek⁴ and increased subsidies for AI infrastructure highlight China's progress.⁵

All of these actions raise the bar to maintain the U.S. AI leadership position and highlight the strategic importance of recent onshoring and investment announcements, from the Stargate Project,⁶ DAMAC Properties,⁷ and others.⁸ to build the infrastructure necessary to continue U.S. progress in AI.

Lumen is Building the Backbone for the AI Economy

Since February 2023, Lumen has been engaged in a far-reaching transformation strategy to cloudify its telecommunications business and adapt the company's national fiber network to meet the needs of hyperscalers and AI-driven enterprises. Lumen now operates the largest ultra-low-loss, intercity fiber network in North America and the single most-peered IP network in the

AI innovation by 2030. See WEF, [Transforming industries with AI: Lessons from China's Journey](#), Jan. 23, 2025.

⁴ See CNBC, [Beijing Embraces DeepSeek to lead AI adoption as it looks for new growth drivers](#), Feb. 21, 2025.

⁵ According to the Diplomat, China has significantly increased its subsidies and incentives for the semiconductor industry, AI development, and data centers, aiming to reduce reliance on U.S.-controlled technology and support domestic and global expansion. See The Diplomat, [China Boosts Semiconductor Subsidies as US tightens restrictions](#), Sept. 28, 2023.

⁶ See Open AI, [Announcing the Stargate Project](#), Jan. 21, 2025.

⁷ See DAMAC, [Trump Announces DAMAC's \\$20 Billion Investment in U.S. Datacenters](#), Jan. 9, 2025.

⁸ Google's recent financial report indicates it plans to spend \$75 billion on advancing AI technologies in 2025. See [Alphabet Announces Fourth Quarter and FY 2024 Results](#). Microsoft has noted it has budgeted approximately \$80 billion for AI-enabled data centers in 2025. See Brad Smith, Microsoft, [The Golden Opportunity for American AI](#), Jan. 3, 2025. Amazon Web Services (AWS) is planning to spend nearly a year of AWS revenue on AI investments in 2025. The company aims to allocate over \$85 billion towards AI, primarily for AWS data centers and AI systems. See The Next Platform, [Amazon Will Spend Nearly a Year of AWS Revenue on AI Investments](#), Feb. 7, 2025.

world,⁹ employing approximately 25,000 team members. The company is rapidly expanding its network, from 12 million inter-city fiber miles in 2022 to an expected 47 million miles in 2028.¹⁰

Lumen is also partnering with U.S.-based Corning to deploy next-generation fiber, which will provide up to four times more capacity over existing and new routes. Lumen also is leveraging photonics innovation for up to twofold greater fiber efficiency. Lumen has the unique ability to scale its network quickly to meet demand, thanks to thousands of miles of multi-conduit infrastructure the company built decades ago along existing fiber routes, enabling it to overpull additional fiber strands through unused or underused conduit. Over the last several months, Lumen also has entered into approximately \$8.5 billion worth of deals with hyperscalers that have chosen Lumen to be their fiber connectivity provider, including Microsoft, Meta, AWS, and Google Cloud.¹¹ Through these partnerships, Lumen will provide them with dramatic increases to their network capacity, thereby supporting their ambitious AI deployment plans. At the same time, Lumen is doing so efficiently, increasing the overall utilization of its network from 57 percent to 70 percent.¹²

In sum, Lumen is expanding its national fiber network to support data center growth, which, in turn, supports U.S. AI deployment and promotes national security and economic competitiveness.

AI Needs Data. Data Needs Data Centers. Data Centers Need to be Connected

In the last six months alone, hyperscalers like Google, Microsoft, Meta, and Amazon Web Services have announced plans to invest, in the aggregate, over \$300 billion to grow their

⁹ See CAIDA ASRank, <https://asrank.caida.org/> (last visited Mar. 13, 2025).

¹⁰ S&P Global Market Intelligence, [Lumen Technologies, Inc. NYSE: LUMN FQ4 2024 Earnings Call Transcripts](#), Feb. 4, 2025.

¹¹ *See id.*

¹² [See id.](#)

respective AI infrastructure, augmented by multi-year investments like the \$100 billion Stargate project.¹³ McKinsey & Company estimates that the global demand for data center capacity could rise at an annual rate of between 19 and 22 percent from 2023 to 2030.¹⁴

Hyperscalers typically need hundreds of fibers between data centers to support the training of AI models, inference and other functions, and each data center generally requires 2-4 diverse fiber routes to ensure redundancy and reliability. Leading fiber manufacturers report that data centers running large language models will require up to five times more optical connectivity than they have today.¹⁵ Data center firms have warned that the fiber network could soon replace GPU supply as the next AI infrastructure bottleneck given the unprecedented fiber connectivity needs for the proliferation of data centers to support AI.¹⁶ Lumen expects fiber demands to further increase as AI data centers decentralize away from major population centers and metro power grids, requiring the construction of new data routes, but also bringing new digital infrastructure to rural communities.¹⁷

To help meet this demand, Lumen expects its total capital expenditures to be in the range of \$4.1 billion to \$4.3 billion in 2025 to accommodate new route construction, fiber overpulls,

¹³ See Fierce Network, [Cloud hyperscalers aren't taking chances on network capacity](#), Feb. 6, 2025; The Next Platform, Amazon Will Spend Nearly a Year of AWS Revenue on [AI Investments](#), Feb. 7, 2025.

¹⁴ McKinsey, [AI power: Expanding data center capacity to meet growing demand](#), Oct. 29, 2024.

¹⁵ See, e.g., [Corning, The Progress Report](#), Dec 2023.

¹⁶ See Fierce Network, [Hyperscalers are buying what Lumen is selling](#), Oct. 30, 2024; Fierce Network, [Cloud hyperscalers aren't taking chances on network capacity](#), Feb. 6, 2025 (quoting industry analyst: “we often speak about electricity shortages due to insatiable data center demand/builds, but high speed and low latency interconnections are equally critical to maintaining performance”).

¹⁷ Data centers in the U.S. tend to be near urban high-tech business centers. As AI applications expand, data center owners will look for large tracts of land for new facilities. See, e.g., Telecompetitor, [AI Drives Demand for Data Centers; Is it a Rural Business Opportunity](#), Sept. 18, 2024.

augmentation of in-line signal amplifiers, and cloudification of Lumen’s telecom services, complemented by similar investments from other carriers and the hyperscalers themselves.¹⁸ However, while U.S. providers like Lumen are investing billions to provide the fiber connectivity necessary to support AI, red tape around permitting and rights-of-way slow the pace of deployment. Efficient permitting and targeted deregulation are needed to remove these deployment barriers, and tax policy that encourages investment in American infrastructure will help expedite deployment.

Policy Recommendations to Support AI Fiber Connectivity

Policymakers have long recognized that restrictive permitting and rights-of-way access hinder cost-effective and timely broadband deployment. These same barriers affect fiber deployment for AI data centers. However, while broadband deployment delays affect specific pockets of consumers, delays in fiber connectivity for AI data centers could pose a risk to the U.S. leadership role in AI. Lumen long has supported proposed policy actions to eliminate these barriers, including “dig once” policies; fair and affordable access to utility poles; coordinating with transportation agencies and railroad operators for rights-of-way access; and streamlining local, state, and regional permitting processes.

In addition to these policy actions, Lumen recommends the following measures to enable the U.S. to solidify and advance its leadership in AI. These measures are aimed at promoting U.S. economic competitiveness and enhancing national security.

- *Establish consistent permitting standards that apply uniformly to all federal, state, and local permitting applications.* The current patchwork of processes creates burdens for

¹⁸ See S&P Global Market Intelligence, [Lumen Technologies, Inc. NYSE: LUMN FQ4 2024 Earnings Call Transcripts](#), Feb. 4, 2025.

companies building critical fiber infrastructure and for government personnel alike, insofar as it fosters inefficiencies for both.

- *In the absence of consistent permitting standards, adopt mutual recognition permitting approval processes for federal, state, and local applications.* Adopting a mutual recognition permitting approval process for federal, state, and local applications will streamline the approval process by reducing redundant reviews. This approach acknowledges state programs with primary regulatory authority, allowing federal agencies to rely on state-approved permits.
- *Adopt cost-based fees.* Local governments often impose excessive fees to generate new revenue streams, which act as hidden taxes on community investments. These fees cause cost overruns and project delays, negatively impacting both companies and consumers. A solution is to implement a flat, cost-based, uniform fee for each permit application regardless of whether filed at the federal, state, or local level.
- *Establish a single point of contact.* Requiring permitting authorities to establish a single point of contact for the permitting process will streamline coordination and expedite applications related to fiber connectivity for AI infrastructure projects. This approach reduces confusion and redundancy, ensuring that all necessary information is centralized and accessible.
- *60-day “shot clock” for all permit applications.* When devising the AI Action Plan, it is crucial to consider additional measures to expedite permitting approvals. For instance, if there are no objections within 30 days, the permit should be automatically approved. Moreover, most applications for upgrading fiber networks to support AI are intended for

deployment in existing conduits, yet they face the same long wait times as new infrastructure projects. Such applications should be subject to a shorter shot clock.

- *Provide for bulk permitting.* The application process should include a bulk permitting process for long-haul projects, requiring only one permit per project.
- *Ensure National Environmental Policy Act (NEPA) categorical exclusions.* Expanding NEPA categorical exclusions to include new fiber routes will speed up the deployment of critical fiber connectivity for AI data centers. Current exemptions only cover expansions to existing infrastructure. Building new fiber routes is well-understood and has minimal environmental impact.
- *Modernize permitting interfaces.* This will provide greater transparency and real-time updates throughout the permitting process.

Lumen also recommends that any task forces or councils proposed in the AI Action Plan to promote AI infrastructure include fiber connectivity providers, and include in their scope advancing fiber deployment permitting reforms in Congress and assisting in resolving any logjams that may arise in interagency coordination efforts geared towards promoting permitting and rights-of-way reforms.

Beyond permitting and rights-of-way reforms, Lumen also supports forward-looking fiscal and regulatory policies that could provide significant incentives to deploy additional AI infrastructure:

- **Encourage Congress to renew the 2017 Tax Cuts and Jobs Act**, especially the provisions on expensing of capex and the business interest deduction.
- **Provide regulatory relief** from federal and state rules that divert limited private capital toward obsolete networks in communities that are already served by modern, digital

networks.

Conclusion

Lumen applauds the Administration for having issued this Request for Information so quickly out of the starting gate. As a leading communications infrastructure provider and the preeminent backbone provider for AI, Lumen welcomes the opportunity to leverage its experience to collaborate with OSTP and NITRD NCO to ensure the AI Action Plan prioritizes concrete, implementable policy actions to promote deployment of fiber infrastructure, which is critical to support the connectivity needs of AI data centers.

Respectfully submitted,

LUMEN TECHNOLOGIES, INC.

By: Melissa H. Mann
SVP, Public Policy and Government Affairs
660 N. Capitol Street, NW, Suite 240
Washington, DC 20001


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